Temporal Video Text Tracking and Distractor Discrimination:

Theoretical Foundations, Data Association Algorithms, and Elimination Cascades

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Abstract

Automated extraction of hardcoded song lyrics from music videos requires transcending static, frame-by-frame scene text detection. In isolated video frames, visual appearance alone cannot disambiguate salient lyrics from persistent background watermarks, broadcast logos, introductory title cards, or incidental scene text. This monograph establishes the theoretical and algorithmic foundations for the temporal tracking and semantic discrimination stage (Phase 1). We formalize Video Text Tracking (VTT) as a constrained multi-target data association problem and establish a two-level hierarchical architecture that decouples spatial footprint continuity from discrete textual state transitions. We analyze four competing data association paradigms: Greedy Spatial Association, Two-Stage Bipartite Matching (ByteTrack-Text [1]), Linear Kalman Motion Modeling (SORT-Text [2]), and Spatio-Temporal Density Clustering (ST-DBSCAN [5, 4]). Furthermore, we formalize the discriminative feature space and derive a hierarchical negative elimination cascade that deterministically purges visual distractors without relying on uncalibrated heuristic filters. Finally, we formulate the standardized evaluation protocol based on CLEAR MOT metrics [6], downstream lyric retrieval scores, and distractor leakage rates.

Keywords: Video text tracking, multi-object tracking, subtitle extraction, ByteTrack, SORT, spatio-temporal clustering, semantic discrimination, music video parsing.

1 Introduction and Mathematical Problem Formulation

1.1 The Single-Frame Ambiguity Dilemma

Given a video stream containing hardcoded text overlays, optical character recognition (OCR) engines operating independently on discrete frames produce spatial bounding geometries and transcribed Unicode strings. However, at any discrete timestamp t, an isolated visual observation provides zero supervisory signal regarding its semantic role within the video production. For instance, within a single frame, a text box transcribing *"We Can Be Alone"* (a salient lyric line), a box transcribing *"Official Music Video"* (a static overlay), and a box transcribing *"Record Label"* (a persistent watermark) exhibit identical geometric topologies, stroke contrasts, and OCR confidence scores.

Relying on hand-crafted spatial assumptions, such as hardcoded lower-third bounding regions of interest, fundamentally fails in modern video productions. In creative music videos

and animations, typography floats dynamically across arbitrary screen coordinates, animates diagonally across camera cuts, or aligns centrally with choreographic action. Recent formulations in general video text spotting, such as GoMatching (He et al., 2024 [7]), and end-to-end vision-language subtitle extraction, such as EVE (Yu et al., 2025 [8]), underscore the necessity of modeling temporal continuity across frames. However, unlike cinematic dialogue subtitles or scene signage, musical lyric typography introduces rapid lexical transitions within stationary bounding zones. Consequently, semantic discrimination cannot be resolved purely in the spatial domain; it must be elevated into the *temporal domain*.

1.2 Formal Mathematical Definition

Let a video stream $\mathcal{V}$ be represented by an ordered sequence of T sampled video frames:

$$\mathcal{V} = \big\{(I_t, \tau_t) \;\big|\; t \in \{1, \dots, T\},\ \tau_1 < \tau_2 < \cdots < \tau_T\big\}, \tag{1}$$

where $I_t \in \mathbb{R}^{H \times W \times 3}$ is the image frame lattice of height H and width W, and $\tau_t \in \mathbb{R}_{\geq 0}$ denotes its exact Presentation Time Stamp (PTS) expressed in milliseconds.

At each frame index t, a text detection and recognition subsystem produces an unordered set of N_t detected text instances:

$$\mathcal{D}_t = \big\{d_{t,i} \;\big|\; i \in \{1, \dots, N_t\}\big\}, \tag{2}$$

where each observation $d_{t,i}$ is defined as a multimodal tuple:

$$d_{t,i} = \big(b_{t,i},\ p_{t,i},\ s_{t,i},\ c_{t,i}\big). \tag{3}$$

The tuple components represent the following spatial, polygonal, textual, and probabilistic attributes:

- $b_{t,i} = [x_{\min}, y_{\min}, x_{\max}, y_{\max}] \in \mathbb{R}^4$: axis-aligned bounding box coordinates,
- $p_{t,i} = [(x_1, y_1), (x_2, y_2), (x_3, y_3), (x_4, y_4)] \in \mathbb{R}^{4 \times 2}$: ordered quadrilateral polygon vertices,
- $s_{t,i} \in \Sigma^*$: normalized transcribed character string over alphabet Σ,
- $c_{t,i} \in [0, 1]$: scalar detection and recognition confidence score.

Definition 1 (Video Text Track). *Video Text Tracking (VTT) partitions the aggregate observation pool $\bigcup_{t=1}^{T} \mathcal{D}_t$ into K mutually disjoint temporal trajectories termed* TextTracks*:*

$$\mathcal{T}_k = \Big\{d_{t,\pi_k(t)} \;\Big|\; t \in \Omega_k \subseteq \{1, \dots, T\}\Big\}, \quad k \in \{1, \dots, K\}, \tag{4}$$

such that every trajectory $\mathcal{T}_k$ corresponds to a unique, temporally coherent visual text entity, subject to the single-instance constraint:

$$\mathcal{T}_j \cap \mathcal{T}_k = \emptyset \quad \forall j \neq k. \tag{5}$$

Once trajectories are formed, the semantic classification mapping f_{cls} assigns each trajectory to a designated taxonomy class:

$$f_{\text{cls}} : \mathcal{T}_k \;\longrightarrow\; \mathcal{C} \in \big\{\text{lyric, channel, song_title, supplementary, scene_text, unknown}\big\}. \tag{6}$$

2 The Two-Level Hierarchical Tracking Architecture

2.1 The Fallacy of Direct Lexical Association

A naive approach to text tracking associates bounding boxes between consecutive frames by measuring textual string similarity, such as normalized Levenshtein distance or character-level Jaro-Winkler scores. In the context of lyric extraction, this formulation introduces a fundamental architectural defect:

Principle 1 (The Fallacy of Direct Lexical Association). *In video text tracking of hard-coded musical lyrics, string edit-distance must not serve as the primary association criterion between consecutive frames. Because visual lyrics undergo instantaneous lexical transitions at phrase boundaries within a stationary visual bounding zone, requiring lexical continuity prematurely severs the temporal trajectory, extinguishing track lifetime and transition frequency metrics.*

When a singer concludes one lyric line and begins the next, the visual text at that screen location changes completely (e.g., from *"Hold my hand"* to *"Into the dark"*). If associations enforce string similarity, the tracking state machine unconditionally terminates the active trajectory and initiates an unrelated track ID. This premature termination has three catastrophic consequences:

1. The total lifespan of the trajectory collapses from the duration of the entire song section down to an isolated, transient two-second snippet.
2. The count of discrete text transitions within that screen zone (n_Δ) is artificially forced to zero for all tracks.
3. Consequently, a genuine lyric line becomes mathematically indistinguishable from a transient scene artifact or a fleeting introductory credit card.

2.2 Decoupled Two-Level Architecture

To resolve this contradiction, the tracking architecture establishes a strict two-level hierarchy that decouples continuous geometric presence from discrete semantic state transitions:

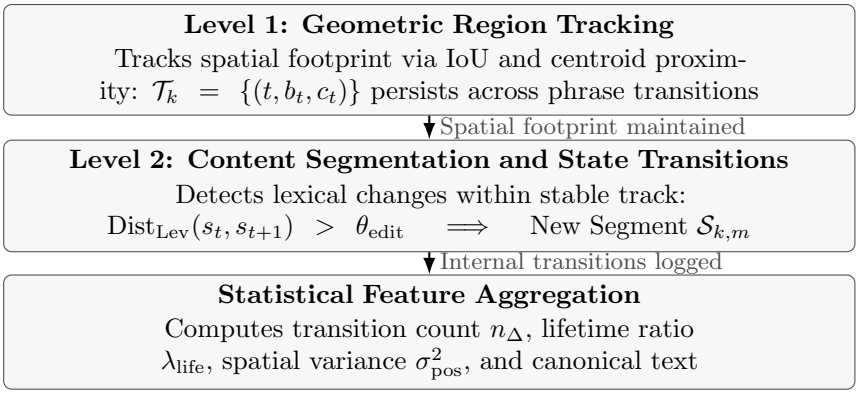


Figure 1: Two-level tracking architecture: geometric region permanence governs track identity, while string distance governs internal sentence state transitions.

The two tracking levels operate as follows:

1. *Level 1 (Spatial Region Tracking)*: Bounding boxes are tracked solely based on spatial overlap (Intersection over Union, IoU) and centroid proximity. As long as text continues to appear within the designated screen region, the track remains continuous and maintains a single unified identity $\mathcal{T}_k$.

2. *Level 2 (Text Change Event Detection)*: Within a stable geometric track $\mathcal{T}_k$, the internal sequence of recognized strings $(s_{t_1}, s_{t_2}, \dots)$ is continuously monitored. A transition event is triggered whenever the normalized string distance exceeds a threshold θ_{edit} for $N_{\text{persist}} \geq 2$ consecutive frames:

$$\text{Event}_{\text{change}}(t) = \mathbf{1}\left[1 - \frac{\text{Levenshtein}(s_t, s_{t-1})}{\max(|s_t|, |s_{t-1}|)} > \theta_{\text{edit}}\right]. \tag{7}$$

   Each stable interval between transition events constitutes a *TextSegment* $\mathcal{S}_{k,m}$, representing an individual lyric phrase.

Under this decoupled formulation, the cumulative transition count:

$$n_\Delta(\mathcal{T}_k) = \sum_{t \in \Omega_k} \text{Event}_{\text{change}}(t) \tag{8}$$

serves as a primary discriminative signature:

- *Persistent watermarks and logos*: Exhibit high lifetime ($\tau_{\text{end}} - \tau_{\text{start}} > 60\%$ of video duration) but zero or near-zero transitions ($n_\Delta \leq 1$).
- *Dynamic lyric tracks*: Exhibit sustained spatial presence with high transition counts ($n_\Delta \geq 4$), with individual segments persisting for human vocal cadences ($1.5\,\text{s} \leq \Delta\tau_{\text{segment}} \leq 8.0\,\text{s}$).

3 Data Association Paradigms in Video Text Tracking

To construct $\mathcal{T}_k$ from frame-level detections, an association algorithm must match existing tracks against candidate detections at frame t. We analyze four competing algorithmic paradigms:

3.1 Algorithm 1: Greedy Spatial IoU & Centroid Matching (Baseline)

The baseline association paradigm matches candidate detections to active tracks greedily using a composite geometric cost matrix. Let $b_A, b_B \in \mathbb{R}^4$ denote two axis-aligned bounding boxes. The Intersection over Union (IoU) is computed as:

$$\text{IoU}(b_A, b_B) = \frac{\text{Area}(b_A \cap b_B)}{\text{Area}(b_A \cup b_B)} = \frac{\max(0, x_2^\cap - x_1^\cap) \cdot \max(0, y_2^\cap - y_1^\cap)}{\text{Area}(b_A) + \text{Area}(b_B) - \text{Area}(b_A \cap b_B)}. \tag{9}$$

To prevent association failure when font sizes vary slightly between lines, the metric incorporates normalized Euclidean centroid distance:

$$D_{\text{centroid}}(b_A, b_B) = \frac{\sqrt{(c_x^A - c_x^B)^2 + (c_y^A - c_y^B)^2}}{\sqrt{W^2 + H^2}}, \tag{10}$$

yielding a combined affinity score:

$$\mathcal{A}(b_A, b_B) = \alpha \cdot \text{IoU}(b_A, b_B) + (1 - \alpha) \cdot \max\left(0, 1 - \frac{D_{\text{centroid}}(b_A, b_B)}{\delta_{\max}}\right). \tag{11}$$

Candidate matches satisfying $\mathcal{A} \geq \theta_{\text{match}}$ are assigned in decreasing order of affinity.

Operational Limitations. Greedy assignment is susceptible to local optima when multiple text lines appear simultaneously in close proximity (such as multi-line choruses or adjacent credits). Furthermore, a single missed detection under illumination shifts immediately flags a track as lost, forcing unnecessary track fragmentation.

3.2 Algorithm 2: Two-Stage Bipartite Association (ByteTrack-Text)

Adapted from the state-of-the-art Multi-Object Tracking framework ByteTrack (Zhang et al., 2022 [1]), this paradigm recognizes that detection confidence fluctuates due to video compression artifacts, motion blur, and semi-transparent background video scenes.

Rather than discarding low-confidence detections below a global threshold τ_{high}, detections $\mathcal{D}_t$ are split into two complementary tiers:

$$\mathcal{D}_{\text{high}} = \left\{ d \in \mathcal{D}_t \;\middle|\; c_d \geq \tau_{\text{high}} \right\}, \qquad \mathcal{D}_{\text{low}} = \left\{ d \in \mathcal{D}_t \;\middle|\; \tau_{\text{low}} \leq c_d < \tau_{\text{high}} \right\}, \tag{12}$$

where typically $\tau_{\text{high}} = 0.70$ and $\tau_{\text{low}} = 0.25$.

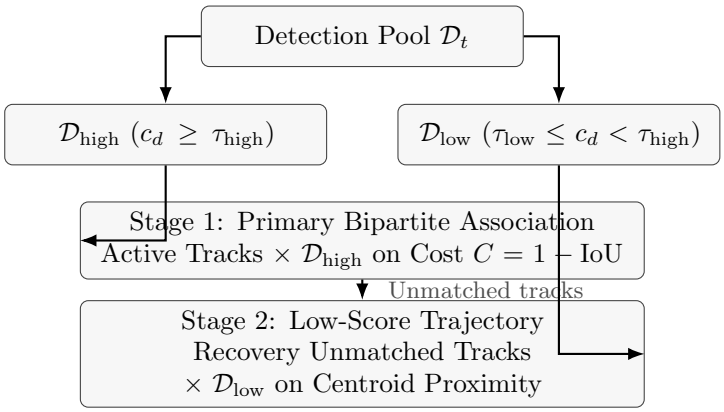


Figure 2: Two-stage association architecture in ByteTrack-Text. Low-confidence detections are utilized exclusively to maintain track continuity across degraded frames.

The matching sequence operates in two structured phases:

1. *Stage 1 (Primary Association)*: Compute the affinity cost matrix:

$$C_{ij}^{(1)} = 1 - \text{IoU}(b_{\text{track},i}, b_{\text{high},j}), \tag{13}$$

and solve the optimal global assignment problem via the Kuhn-Munkres (Hungarian) algorithm:

$$\mathbf{X}^* = \arg\min_{\mathbf{X}} \sum_i \sum_j C_{ij}^{(1)} X_{ij} \quad \text{s.t.} \quad \sum_j X_{ij} \leq 1,\ \sum_i X_{ij} \leq 1,\ X_{ij} \in \{0,1\}. \tag{14}$$

Pairs with IoU $< \theta_{\text{IoU}}$ are discarded.

2. *Stage 2 (Low-Confidence Recovery)*: Active tracks remaining unmatched after Stage 1 are paired against $\mathcal{D}_{\text{low}}$ using a relaxed centroid distance matrix. This recovers tracks during brief illumination dips or motion blur without permitting low-confidence false positives to initiate spurious new tracks.

3. *Trajectory State Management*: Unmatched high-confidence detections initiate tentative tracks; tracks unobserved for $\Delta\tau > \tau_{\text{max_gap}}$ (e.g., 2000 ms) are finalized and archived.

3.3 Algorithm 3: Linear Kalman Motion Modeling (SORT-Text)

When video typography is dynamic (such as scrolling end-credits, karaoke bounce animations, or kinetic title overlays), static IoU between consecutive frames drops to zero if text motion exceeds the stroke height.

The Simple Online and Realtime Tracking (SORT) framework (Bewley et al., 2016 [2]; Yu et al., 2019 [3]) models each text instance's spatial trajectory as a linear dynamical system:

$$\mathbf{x}_k = \mathbf{F}\mathbf{x}_{k-1} + \mathbf{w}_k, \qquad \mathbf{z}_k = \mathbf{H}\mathbf{x}_k + \mathbf{v}_k, \tag{15}$$

where the state vector is parameterized by the bounding box center (u, v), scale/area $s = w \cdot h$, aspect ratio $r = w/h$, and their continuous velocities:

$$\mathbf{x} = \big[u,\, v,\, s,\, r,\, \dot{u},\, \dot{v},\, \dot{s}\big]^\top \in \mathbb{R}^7. \tag{16}$$

Assuming constant velocity, the state transition matrix $\mathbf{F}$ is:

$$\mathbf{F} = \begin{bmatrix} \mathbf{I}_3 & \mathbf{0} & \Delta t \cdot \mathbf{I}_3 \\ \mathbf{0} & 1 & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{I}_3 \end{bmatrix}, \qquad \mathbf{H} = \big[\mathbf{I}_4 \quad \mathbf{0}_{4\times 3}\big]. \tag{17}$$

At frame t, the Kalman filter predicts the prior distribution $\hat{\mathbf{x}}_{t|t-1} = \mathbf{F}\hat{\mathbf{x}}_{t-1}$ and error covariance $\mathbf{P}_{t|t-1} = \mathbf{F}\mathbf{P}_{t-1}\mathbf{F}^\top + \mathbf{Q}$. The association cost matrix is evaluated between predicted boxes $\hat{b}_{t|t-1}$ and measured detections b_t. Once matched, the posterior estimate is updated:

$$\mathbf{K}_t = \mathbf{P}_{t|t-1}\mathbf{H}^\top \Big(\mathbf{H}\mathbf{P}_{t|t-1}\mathbf{H}^\top + \mathbf{R}\Big)^{-1}, \qquad \hat{\mathbf{x}}_t = \hat{\mathbf{x}}_{t|t-1} + \mathbf{K}_t\big(\mathbf{z}_t - \mathbf{H}\hat{\mathbf{x}}_{t|t-1}\big)\,. \tag{18}$$

This allows SORT-Text to track scrolling credits and moving text smoothly across significant frame-to-frame displacements.

3.4 Algorithm 4: Spatio-Temporal Tube Clustering (ST-DBSCAN)

Unlike online tracking algorithms that process frames sequentially and are vulnerable to irreversible error accumulation, offline video processing permits batch spatio-temporal clustering (Sakaguchi et al., 2020 [4]; Ester et al., 1996 [5]).

All detections across the entire video timeline are embedded into a normalized 4-dimensional spatio-temporal metric space $\mathcal{M} = \mathbb{R}^4$:

$$\mathbf{p}_i = \left[\frac{c_x^{(i)}}{W},\, \frac{c_y^{(i)}}{H},\, \frac{\tau^{(i)}}{\tau_{\text{video}}},\, \log\!\left(\frac{\text{Area}(b_i)}{W \cdot H}\right)\right]^\top \in \mathbb{R}^4. \tag{19}$$

The distance between two observations d_i, d_j is evaluated via a non-isotropic weighted metric:

$$\mathcal{D}(\mathbf{p}_i, \mathbf{p}_j) = \sqrt{w_s\left(\left(\frac{c_x^i - c_x^j}{W}\right)^2 + \left(\frac{c_y^i - c_y^j}{H}\right)^2\right) + w_t\left(\frac{|\tau^i - \tau^j|}{\tau_{\text{max_gap}}}\right)^2 + w_a\left(\log\frac{A_i}{A_j}\right)^2}. \tag{20}$$

Density-Based Spatial Clustering of Applications with Noise (DBSCAN) identifies core points possessing at least MinPts neighbors within an ε-neighborhood:

$$\mathcal{N}_\varepsilon(\mathbf{p}) = \big\{\mathbf{q} \in \mathcal{M} \;\big|\; \mathcal{D}(\mathbf{p}, \mathbf{q}) \le \varepsilon\big\}, \qquad |\mathcal{N}_\varepsilon(\mathbf{p})| \ge \text{MinPts}\,. \tag{21}$$

Connected density components form continuous *text tubes*. Because ST-DBSCAN operates globally, it is completely immune to frame-order biases and cleanly bridges multi-second temporal pauses where a singer pauses during an instrumental break before resuming lyrics in the exact same spatial zone.

4 Discriminative Classification and the Elimination Cascade

Once text trajectories $\mathcal{T}_k$ are assembled, the classification engine maps each trajectory to its categorical label $\mathcal{C}$.

4.1 Multi-Dimensional Statistical Feature Representation

For each trajectory $\mathcal{T}_k$, a compact feature vector $\boldsymbol{\phi}(\mathcal{T}_k)$ is computed from the aggregated occurrences:

Table 1: Taxonomy of statistical features extracted per TextTrack trajectory.

Category	Feature Symbol	Mathematical Definition / Semantic Description
Temporal	λ_{life}	Lifetime ratio: $(\tau_{\text{end}} - \tau_{\text{start}})/\tau_{\text{duration}}$
Temporal	σ_{start}	Relative start time: $\tau_{\text{start}}/\tau_{\text{duration}}$
Temporal	n_Δ	Number of discrete text transitions (7)
Temporal	$\bar{\tau}_{\text{seg}}$	Mean duration of internal segments: $\frac{1}{M}\sum_{m=1}^{M}(\tau_{\text{end}}^{(m)} - \tau_{\text{start}}^{(m)})$
Spatial	$(\bar{c}_x, \bar{c}_y)$	Mean centroid normalized to $[0,1] \times [0,1]$
Spatial	σ^2_{pos}	Normalized spatial trajectory variance: $\frac{1}{\lvert\mathcal{T}_k\rvert}\sum_t \lVert c_t - \bar{c}\rVert_2^2/(W^2 + H^2)$
Spatial	ρ_{aspect}	Bounding box aspect ratio: $\bar{w}/\bar{h}$
Spatial	α_{area}	Bounding box area ratio: $(\bar{w} \cdot \bar{h})/(W \cdot H)$
Spatial	χ_{corner}	Binary indicator: bounding box resides within 5% screen margin
Semantic	$\text{Sim}_{\text{title}}$	Maximum token/Levenshtein similarity to video title metadata
Semantic	$\text{Sim}_{\text{channel}}$	Maximum token/Levenshtein similarity to artist/channel metadata
Semantic	χ_{keyword}	Binary flag: presence of UI/credit keywords (`subscribe`, `directed by`)
Semantic	$\bar{c}_{\text{conf}}$	Mean OCR detection confidence across all occurrences

4.2 The Hierarchical Negative Elimination Cascade

Empirical findings in video text parsing indicate that positive rule matching for lyrics yields high false positive rates due to the sheer diversity of musical typography. In contrast, non-lyric distractors satisfy highly constrained physical and operational invariants.

The system implements a *hierarchical negative elimination cascade*, where candidate tracks must survive four consecutive rejection gates before acceptance:

Gate 1: Persistent Channel and Watermark Rejection. Channel watermarks and broadcast station logos exhibit prolonged static presence with zero lexical variation:

$$\text{Reject}_{\text{channel}}(\mathcal{T}_k) = \mathbf{1}\Big[\big(\lambda_{\text{life}} \geq 0.55 \ \wedge \ n_\Delta \leq 1\big) \ \vee \ \big(\text{Sim}_{\text{channel}} \geq 0.70 \ \wedge \ \lambda_{\text{life}} \geq 0.15\big)\Big]. \quad (22)$$

Matches are classified as `channel` (Decision: *Reject*).

Gate 2: Introductory and Terminal Title Cards. Title cards containing song names, composers, and featured artists appear strictly within the initial or terminal temporal boundaries:

$$\text{Reject}_{\text{title}}(\mathcal{T}_k) = \mathbf{1}\Big[\sigma_{\text{start}} \leq 0.08 \ \wedge \ (\tau_{\text{end}} - \tau_{\text{start}} \leq 12\,\text{s}) \ \wedge \ \big(\text{Sim}_{\text{title}} \geq 0.75 \ \vee \ n_\Delta = 0\big)\Big]. \quad (23)$$

Matches are classified as `song_title` (Decision: *Reject*).

Gate 3: Margin User Interface and Supplementary Text. Social media handles, subscribe prompts, and copyright credits reside in extreme boundary zones or contain explicit regulatory keywords:

$$\text{Reject}_{\text{supp}}(\mathcal{T}_k) = \mathbf{1}\Big[\chi_{\text{keyword}} = 1 \ \vee \ \big(\chi_{\text{corner}} = 1 \ \wedge \ \rho_{\text{aspect}} \leq 2.0\big)\Big]. \tag{24}$$

Matches are classified as `supplementary` (Decision: *Reject*).

Gate 4: Dynamic Diegetic Scene Text. Natural text embedded on physical objects in the video scene (posters, street signs, vehicle branding) undergoes erratic translation driven by camera pan and perspective shifts:

$$\text{Reject}_{\text{scene}}(\mathcal{T}_k) = \mathbf{1}\Big[\sigma^2_{\text{pos}} \geq 0.08 \ \vee \ \big(\rho_{\text{aspect}} < 1.0 \ \wedge \ n_{\Delta} = 0\big)\Big]. \tag{25}$$

Matches are classified as `scene_text` (Decision: *Reject*).

Gate 5: Salient Lyric Acceptance Invariants. Tracks surviving Gates 1 through 4 are evaluated for lyric invariants. A legitimate lyric line occupies human vocal duration ($1.5\,\text{s} \leq \bar{\tau}_{\text{seg}} \leq 8.5\,\text{s}$), maintains a horizontal aspect ratio ($\rho_{\text{aspect}} \geq 1.5$), and exhibits high confidence ($\bar{c}_{\text{conf}} \geq 0.65$):

$$f_{\text{cls}}(\mathcal{T}_k) = \begin{cases} \texttt{lyric} \quad \text{(Accept)}, & \text{if } \bigwedge_{j=1}^{4} (\text{Reject}_j(\mathcal{T}_k) = 0) \text{ and within lyric bounds}, \\ \texttt{unknown} \quad \text{(Review)}, & \text{otherwise (routed to human-in-the-loop audit)}. \end{cases} \tag{26}$$

5 Quantitative Evaluation Protocols and Benchmarking Standards

To compare association algorithms objectively, the evaluation protocol establishes standardized metrics measuring tracking stability and semantic purity.

5.1 Multi-Object Tracking Metrics (CLEAR MOT)

Following the standard CLEAR MOT protocol (Bernardin & Stiefelhagen, 2008 [6]):

1. *Multiple Object Tracking Accuracy (MOTA)*: Measures cumulative tracking errors incorporating false negatives, false positives, and identity switches:

$$\text{MOTA} = 1 - \frac{\sum_t \left(\text{FN}_t + \text{FP}_t + \text{IDSW}_t\right)}{\sum_t \text{GT}_t}, \tag{27}$$

   where $\text{FN}_t, \text{FP}_t, \text{GT}_t$ denote false negatives, false positives, and ground truth instances, and IDSW_t denotes *Identity Switches* (instances where a previously tracked text changes its assigned trajectory ID).

2. *Identification F1-Score (IDF1)*: Measures the ratio of correctly identified trajectory associations over ground truth and computed detections:

$$\text{IDF1} = \frac{2\,\text{IDTP}}{2\,\text{IDTP} + \text{IDFP} + \text{IDFN}}. \tag{28}$$

5.2 Downstream Semantic Metrics

Tracking performance directly influences lyric classification:

1. *Lyric Precision, Recall, and F1-Score*: Evaluates the final subset of tracks accepted as `lyric` against manually confirmed ground truth tracks.

2. *Distractor Leakage Rate (*DLR*)*: The primary safety-critical metric of the Phase 1 pipeline. Measures the percentage of true distractor tracks (watermarks, logos, titles) that bypass all rejection gates and leak into the accepted lyric pool:

$$\text{DLR} = \frac{|\mathcal{T}_{\text{accepted}} \cap \mathcal{T}^*_{\text{distractor}}|}{|\mathcal{T}^*_{\text{distractor}}|} \times 100\%. \tag{29}$$

   The formal operational threshold for Phase 1 requires $\text{DLR} = 0.0\%$.

6 Algorithmic Synthesis and Theoretical Comparison

Table 2: Comparative analysis of tracking association paradigms for video text.

Property	**Greedy IoU**	**ByteTrack-Text**	**SORT-Text**	**ST-DBSCAN**
Paradigm	Online greedy	Online two-stage Hungarian	Online Kalman filter + Hungarian	Offline batch spatio-temporal clustering
Time Complexity	$\mathcal{O}(T \cdot MN)$	$\mathcal{O}(T \cdot MN \log(M+N))$	$\mathcal{O}(T \cdot MN \log(M+N))$	$\mathcal{O}(N_{\text{total}} \log N_{\text{total}})$
Memory Footprint	$\mathcal{O}(M_{\text{active}})$	$\mathcal{O}(M_{\text{active}})$	$\mathcal{O}(M_{\text{active}} \cdot \dim(\mathbf{x}))$	$\mathcal{O}(N_{\text{total}} \cdot \dim(\mathbf{p}))$
ID Switch Resistance	Low (severs on single missing frame)	High (rescues via second-tier association)	Moderate (sensitive to non-linear camera jerks)	Immune to temporal pauses
Dynamic Typography	Poor (static IoU collapses to zero)	Moderate (handles small displacements)	Strong (Kalman velocity propagation)	Strong (spatio-temporal clustering)
Implementation Complexity	Minimal (~ 100 LOC)	Low (~ 250 LOC)	Moderate (~ 350 LOC)	Moderate (~ 200 LOC)

7 Conclusion

This monograph establishes the formal theoretical substrate for Phase 1: Video Text Tracking and Semantic Distractor Discrimination. By decoupling spatial region continuity from discrete textual state transitions, the two-level tracking architecture transforms temporal duration and transition frequency into definitive discriminative features. The comparative analysis among Greedy IoU, ByteTrack, SORT, and ST-DBSCAN defines the benchmark space for evaluating identity permanence and tracking robustness, while the hierarchical negative elimination cascade provides a rigorous, zero-leakage foundation for subsequent lyric synchronization and LRC export.

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